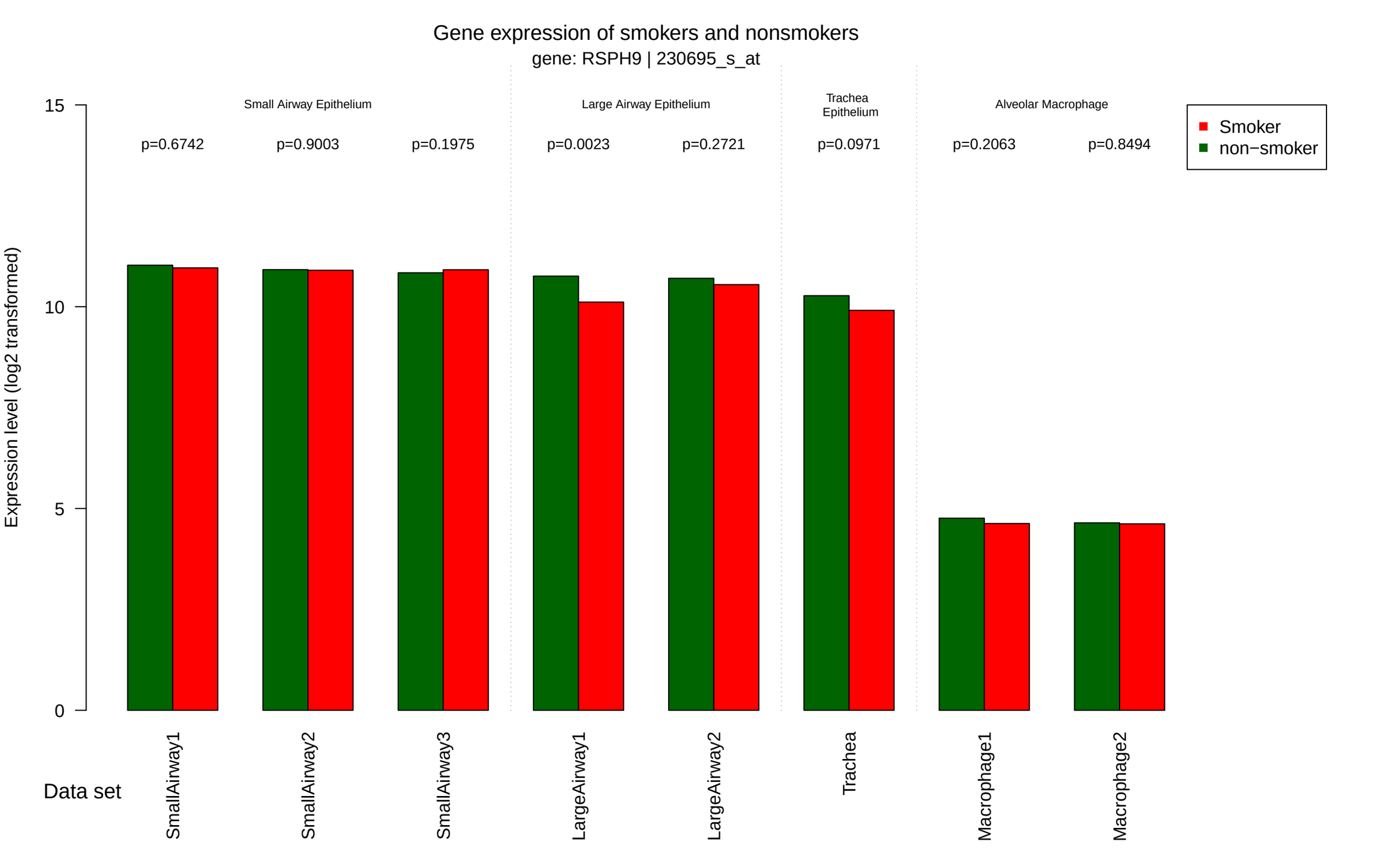
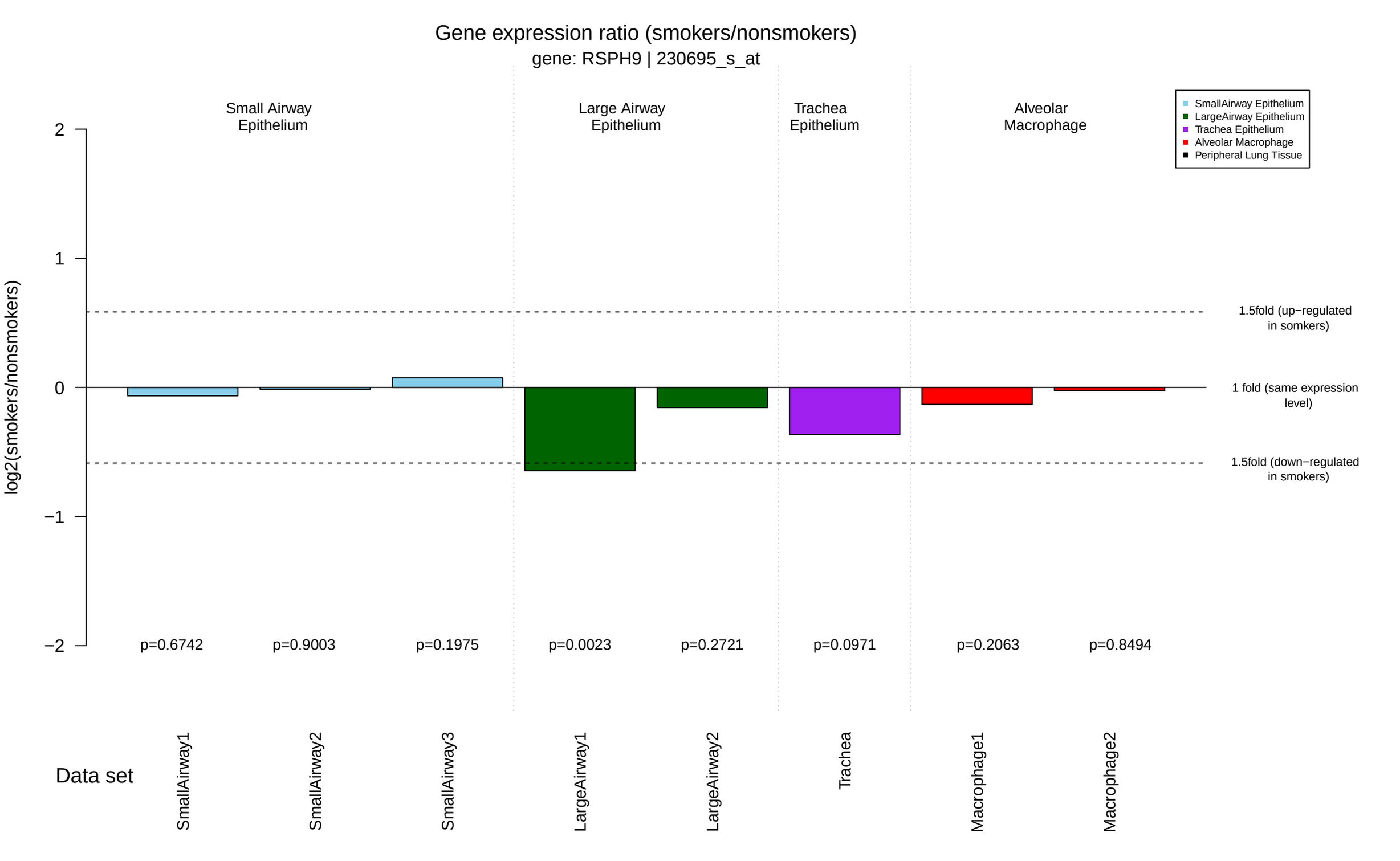
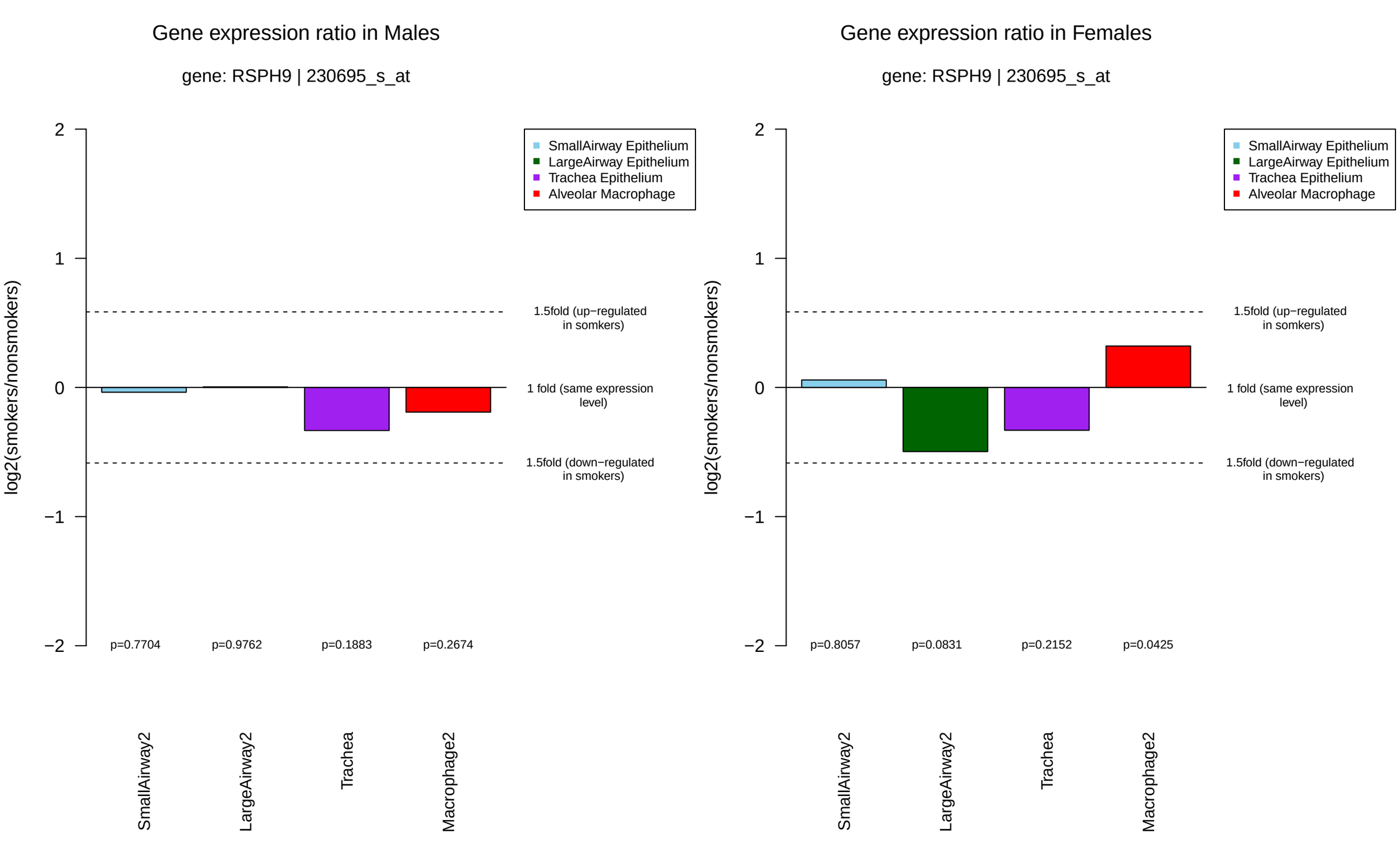
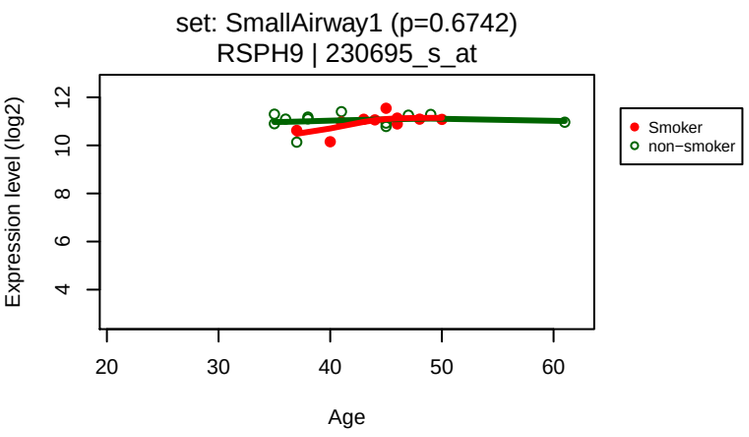
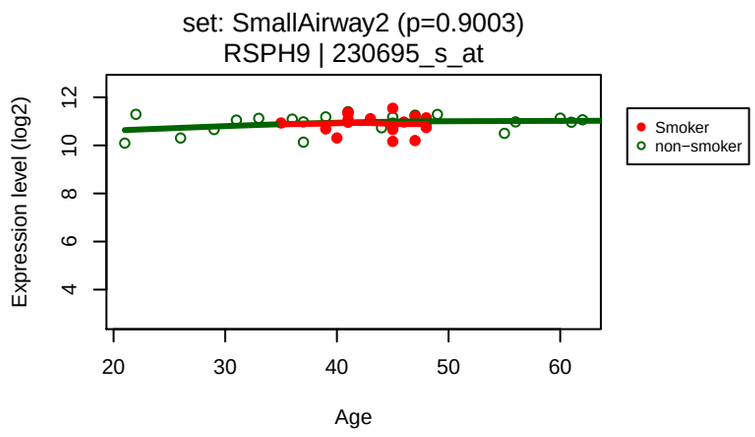
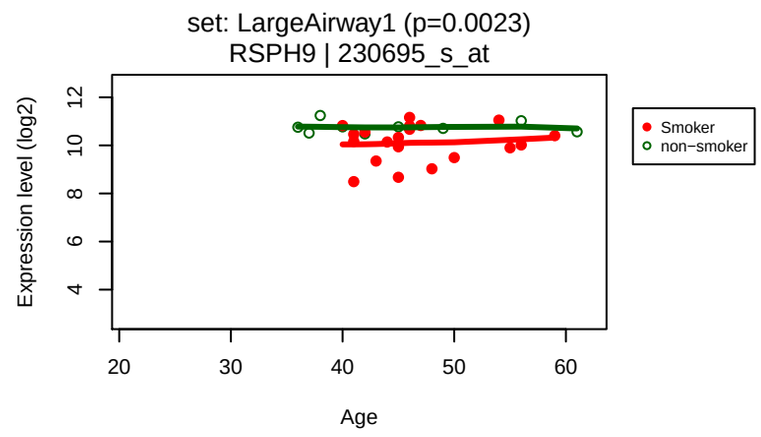

Gene expression of smokers and nonsmokers
gene: RSPH9 | 230695_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6742
p=0.9003
p=0.1975
p=0.0023
p=0.2721
p=0.0971
p=0.2063
p=0.8494
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

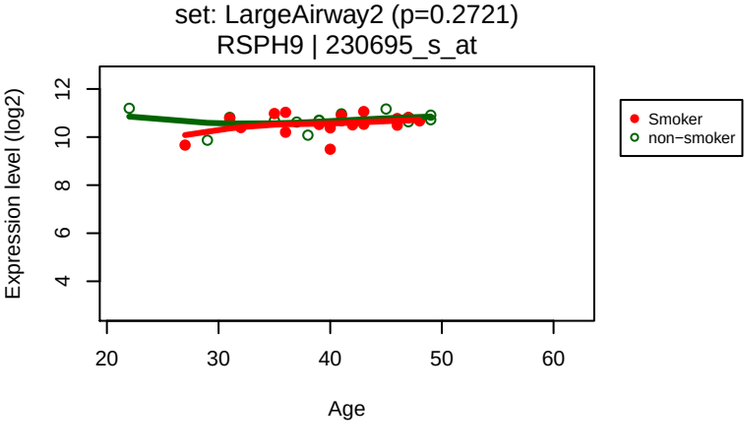

Gene expression ratio (smokers/nonsmokers)
gene: RSPH9 | 230695_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6742
p=0.9003
p=0.1975
p=0.0023
p=0.2721
p=0.0971
p=0.2063
p=0.8494
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

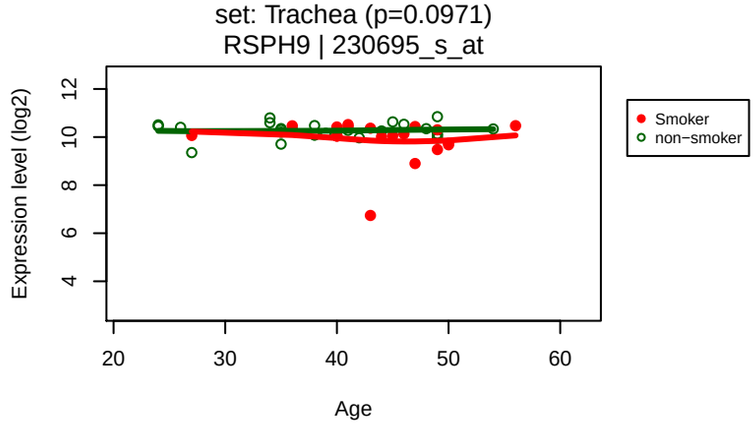

Gene expression ratio in Males
gene: RSPH9 | 230695_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7704
p=0.9762
p=0.1883
p=0.2674
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RSPH9 | 230695_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8057
p=0.0831
p=0.2152
p=0.0425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

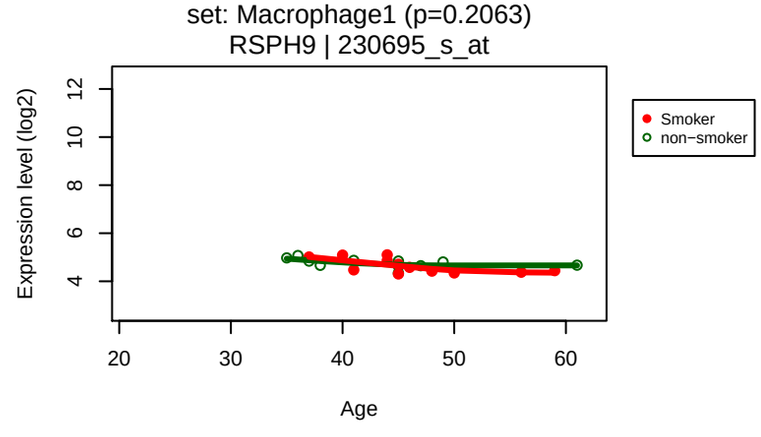

set: SmallAirway1 (p=0.6742)
RSPH9 | 230695_s_at
Expression level (log2)
Age
Smoker
non−smoker

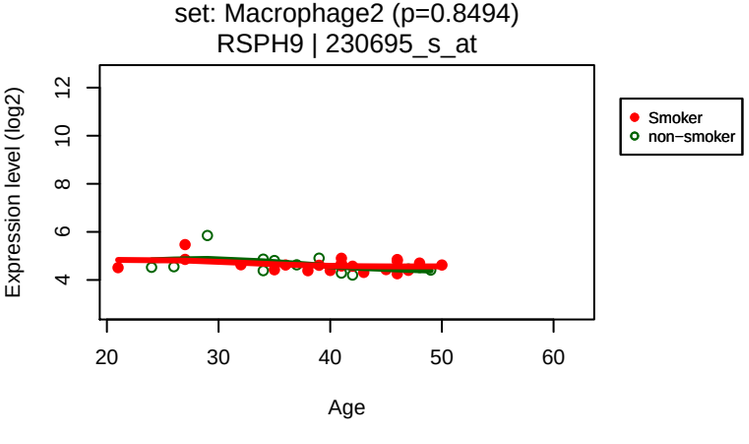

set: SmallAirway2 (p=0.9003)
RSPH9 | 230695_s_at
Expression level (log2)
Age
Smoker
non−smoker

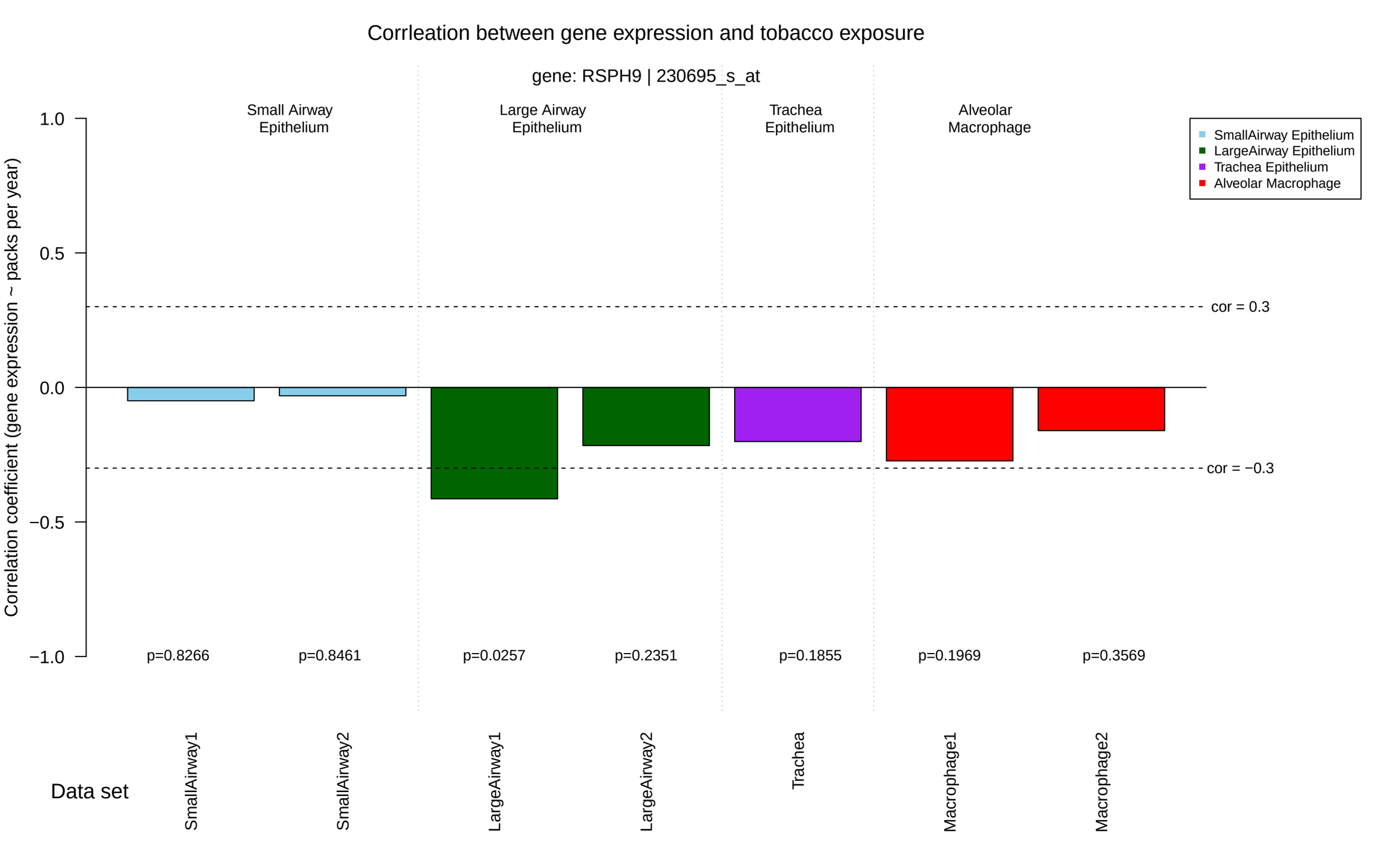

set: LargeAirway1 (p=0.0023)
RSPH9 | 230695_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2721)
RSPH9 | 230695_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0971)
RSPH9 | 230695_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2063)
RSPH9 | 230695_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8494)
RSPH9 | 230695_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RSPH9 | 230695_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8266
p=0.8461
p=0.0257
p=0.2351
p=0.1855
p=0.1969
p=0.3569
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2